7.3 Rectangles, Rhombi, and Squares

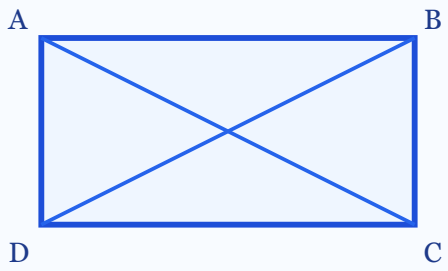
Rectangles, rhombi, and squares are all parallelograms, so they inherit every parallelogram property (opposite sides parallel and congruent, opposite angles congruent, consecutive angles supplementary, diagonals bisect each other) — plus one extra rule each. A **rectangle** has four right angles, which forces its diagonals to be *congruent*. A **rhombus** has four congruent sides, which forces its diagonals to be *perpendicular* and to *bisect the vertex angles*. A **square** is both a rectangle and a rhombus, so it has all of these properties at once. Because a rhombus's diagonals are perpendicular and bisect each other, each half-diagonal pair and a side form a right triangle — the Pythagorean Theorem connects diagonal lengths to side lengths. This worksheet uses all of these relationships, often in combination.

Examples

Example 1: Rectangle diagonals are congruent

Rectangle $ABCD$ has diagonal $AC = 16$. Find BD, and find AO, BO, CO, DO where O is the intersection of the diagonals.

Solution



A rectangle's diagonals are congruent and bisect each other.

Rectangle diagonals are always congruent: $BD = AC = 16$.

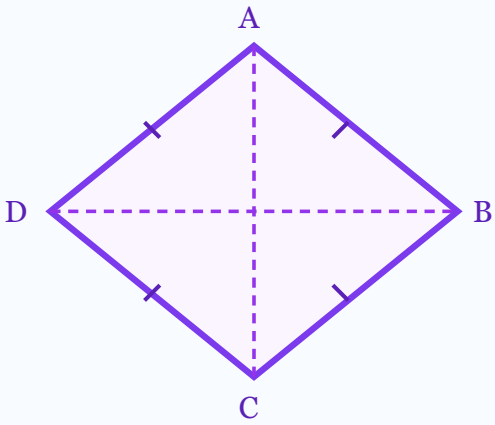
Diagonals of any parallelogram bisect each other, so O is the midpoint of both diagonals: $AO = OC = \frac{16}{2} = 8$ and $BO = OD = \frac{16}{2} = 8$.

Answer: $BD = 16$; all four segments $AO = BO = CO = DO = 8$.

Example 2: Rhombus perimeter and diagonal

A rhombus has perimeter 40. One diagonal is 12. Find the side length and the other diagonal.

Solution



Diagonals of a rhombus are perpendicular bisectors of each other, forming right triangles with the sides.

Side length: $\frac{40}{4} = 10$.

The diagonals are perpendicular and bisect each other, so half of the given diagonal (6) and half of the unknown diagonal form the legs of a right triangle with the side (10) as hypotenuse:

$6^2 + h^2 = 10^2 \Rightarrow 36 + h^2 = 100 \Rightarrow h^2 = 64 \Rightarrow h = 8$.

Full diagonal $= 2(8) = 16$.

Answer: side $= 10$; other diagonal $= 16$.

Example 3: Rhombus side from both diagonals

A rhombus has diagonals 24 and 10. Find the side length.

Solution

The diagonals bisect each other at right angles, giving half-diagonals 12 and 5.

Side $= \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$.

Answer: 13 (this is the 5-12-13 Pythagorean triple).

Example 4: Square diagonal and area

A square has side 5. Find the diagonal, and use it to find the area a second way.

Solution

Diagonal: $d = 5\sqrt{2}$ (the diagonal of a square is side $\times \sqrt{2}$, from the 45-45-90 relationship).

Method 1 (side): Area $= 5^2 = 25$.

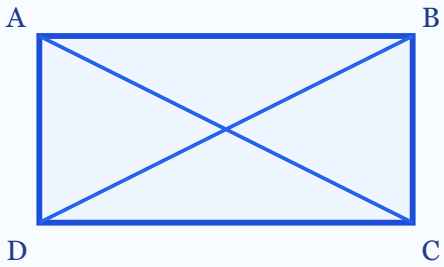
Method 2 (diagonal): Area $= \dfrac{d^2}{2} = \dfrac{(5\sqrt{2})^2}{2} = \dfrac{50}{2} = 25$. Same answer. ✓

Answer: diagonal $= 5\sqrt{2}$; area $= 25$.

Example 5: Rectangle diagonal angle

In rectangle $ABCD$, diagonal $\overline{AC}$ is drawn, and $m\angle BAC = 32°$. Find $m\angle ACD$.

Solution



$\overline{AB} \parallel \overline{DC}$; diagonal $\overline{AC}$ is the transversal.

$\overline{AB} \parallel \overline{DC}$, and $\overline{AC}$ is a transversal crossing both.

$\angle BAC$ and $\angle ACD$ are alternate interior angles, so they are congruent: $m\angle ACD = 32°$.

Answer: $32°$.

Example 6: Rhombus diagonals bisect the vertex angles

In rhombus $ABCD$, $m\angle A = 70°$. Diagonal $\overline{AC}$ is drawn. Find $m\angle BAC$ and $m\angle DAC$, then find $m\angle B$.

Solution

A rhombus's diagonal bisects the vertex angle it passes through, so it splits $\angle A$ into two equal halves:

$$m\angle BAC = m\angle DAC = \frac{70°}{2} = 35°.$$

For the full angle: a rhombus is a parallelogram, so consecutive angles are supplementary:

$m\angle B = 180° - 70° = 110°$.

Answer: $\angle BAC = \angle DAC = 35°$; $\angle B = 110°$.

Example 7: Square from its diagonal

A square has diagonal $10\sqrt{2}$. Find the side length and the perimeter.

Solution

Since $d = s\sqrt{2}$: $10\sqrt{2} = s\sqrt{2} \Rightarrow s = 10$.

Perimeter $= 4(10) = 40$.

Answer: side $= 10$; perimeter $= 40$.

Example 8: Identify the figure from its properties

A parallelogram has perpendicular diagonals and also has a right angle. What figure must it be, and why?

Solution

Perpendicular diagonals in a parallelogram force all four sides to be congruent — that makes it a **rhombus**.

A rhombus with one right angle has all four angles equal to $90°$ (opposite angles congruent, consecutive angles supplementary) — that makes it a **rectangle** too.

A figure that is both a rhombus and a rectangle is a **square**.

Example 9: Rectangle diagonal from side lengths

A rectangle has sides 9 and 12. Find the length of a diagonal.

Solution

A rectangle's diagonal is the hypotenuse of a right triangle formed with the two sides:

$d = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15.$

Answer: 15.

Example 10: Rhombus — side and one diagonal given, find the other diagonal and the area

A rhombus has side 17 and one diagonal 16. Find the other diagonal and the area of the rhombus.

Solution

Half of the known diagonal: $\frac{16}{2} = 8.$

This is a leg of a right triangle with hypotenuse 17 (the side): $8^2 + h^2 = 17^2 \Rightarrow 64 + h^2 = 289 \Rightarrow h^2 = 225 \Rightarrow h = 15.$

Full other diagonal $= 2(15) = 30.$

Area of a rhombus $= \frac{1}{2} d_1 d_2 = \frac{1}{2}(16)(30) = 240.$

Answer: other diagonal $= 30$; area $= 240$.

Example 11: Application — a square window brace

A square window has a diagonal support brace measuring $8\sqrt{2}$ ft. Find the perimeter and area of the window.

Solution

Diagonal $= s\sqrt{2} = 8\sqrt{2} \Rightarrow s = 8$ ft.

Perimeter $= 4(8) = 32$ ft.

Area $= 8^2 = 64$ ft^2.

Answer: perimeter $= 32$ ft; area $= 64$ ft^2.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Rectangle

Diagonal $AC = 22$. Find BD.

Answer: **Verification:**

Problem 2: Rhombus

Perimeter 60. Each side?

Answer: **Verification:**

Problem 3: Rhombus

Diagonals 10 and 24. Side length?

Answer: **Verification:**

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Problem 4: Square

Side 7. Diagonal?

Answer: **Verification:**

Problem 5: Square

Diagonal $6\sqrt{2}$. Side?

Answer: **Verification:**

Problem 6: Rhombus angles

One angle 55°. Adjacent angle?

Answer: **Verification:**

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Problem 7: Rectangle

Sides 5 and 12. Diagonal?

Answer: **Verification:**

Problem 8: Identify

Four congruent sides and four right angles. Name the figure.

Answer: **Verification:**

Problem 9: Rhombus

Diagonals 8 and 6. Side?

Answer: **Verification:**

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Problem 10: Rectangle

Do diagonals of a rectangle always bisect its angles?

Answer: **Verification:**

Problem 11: Square

Perimeter 36. Diagonal?

Answer: **Verification:**

Problem 12: Rhombus

Side 25, one diagonal 14. Find the other diagonal.

Answer: **Verification:**

Problem 13: Property

Diagonals of a rhombus are always...

Answer: **Verification:**

Problem 14: Challenge

A rhombus has diagonals in ratio $3:4$ and area 96. Find both diagonal lengths and the side length.

Answer: **Verification:**

Problem 15: Challenge

The diagonal of a square is 4 cm longer than its side. Find the side length to the nearest tenth of a centimeter.

Answer: **Verification:**

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✓ Answer Key

1. 22
2. 15
3. 13
4. $7\sqrt{2}$
5. 6
6. 125°
7. 13
8. Square
9. 5
10. No (only if it's a square)
11. $9\sqrt{2}$
12. 48
13. Perpendicular bisectors of each other, and they bisect the vertex angles
14. Diagonals 12 and 16; side 10
15. ≈ 9.7 cm

Great work — keep practising!